

Oliver Michel — Research Statement

I am a networking and systems researcher working on improving the performance and scalability of *emerging networked applications*, in the face of their rapid growth, complexity, and associated resource overheads. Due to this complexity, achieving this goal warrants a holistic cross-layer approach where I cover the entire stack, from the physical layer to the higher-layer intricacies of rich, user-facing applications. My research takes a unique measurement-driven approach that emphasizes on understanding the operation and performance of applications *in the wild* and *across layers* by designing measurement methodologies and building measurement systems before crafting practical measurement studies that provide unique insight into actual deployments of large-scale systems. I use these insights to address problems across the stack by designing and building systems that leverage advances in network programmability to improve Quality of Experience (QoE) for users while simplifying and reducing overheads of running these systems for operators.

Research Vision. Many aspects of our lives, including in healthcare, education, and e-commerce, rely on increasingly sophisticated networked applications, like video conferencing and virtual reality, while the applications themselves critically depend on the performance of the underlying network. Operating and scaling these applications is hard as they evolved into *complex and resource-intensive systems* associated with *stringent performance and predictability requirements*. Not meeting these requirements affects all stakeholders: Users may experience poor QoE, such as choppy video calls, while operators are faced with significant revenue loss for just milliseconds of added latency or seconds of downtime. The ultimate goal of my research is to solve this conundrum and to enable scalable and high-performance modern networked applications. Toward this goal, my work is driven by two high-level challenges, which I outline next.

Closing the Semantic Gap between Applications and Networks. My measurement studies have revealed a common theme: the lack of transparency and coordination between applications and networks. Autonomous, closed-loop network control promises to enable such coordination with the goal of optimizing networks for modern applications. Existing network-level monitoring systems, however, do not provide the required application-level insight to realize this vision. This results in networks that do not understand the relevant metrics and performance objectives of applications and, consequently, are unable to optimize themselves for the applications' needs. Conversely, applications often see the network through a clouded window and are unable to correctly interpret artifacts from the network (e.g., sudden latency spikes). To solve this problem, I design measurement methodologies and systems and craft measurement studies that *provide the required application-level insights* to enable closed-loop network control at scale [3, 4, 6, 7].

Boosting Application Performance and Scalability. With surging demand, growing link capacities, and the end of Moore's law, it is increasingly difficult to efficiently run and scale networked systems, posing a significant problem for operators and ultimately compromising QoE. This affects both the class of emerging applications described above as well as lower-level, packet-processing systems such as firewalls or routers. In my work, I show through building systems how leveraging application-level monitoring data and insights inside the network can facilitate significant performance and quality-of-service improvements for a wide range of applications. The resulting networked systems ultimately *tie the network and applications* and use a mix of traditional optimization techniques within novel system architectures [1] as well as modern network programmability using accelerators and programmable switches [2, 3, 5].

My measurement-driven approach tackling these two fundamental challenges of modern networked systems is fruitful as it enables a natural workflow where bridging the semantics of applications and networks reveals cross-layer optimization opportunities, leading to more powerful, reliable, and scalable systems. My work has provided important results, brought new insights, and improved system and application performance. More broadly, however, it has had practical impact on the networking community: methods and artifacts of my work have been adopted by industry, e.g., at Arista Networks, and are widely used by Universities, e.g., at Columbia University, Purdue University, and the University of California Santa Barbara.

Outline – My research spans three areas. First, my work on *video conferencing* shows how to measure and optimize complex, emerging applications. Second, my work on *Linux networking* takes a more general approach and explores how to accelerate a wide range of network functions. Third, my work on *network monitoring and control* combines learnings from measurement and system design, enabling closed-loop network control.

Performance and Scalability of Video-Conferencing Applications

My work on video conferencing started with enabling visibility into the increasingly important area of real-time communications, which also includes, for example, live streaming and cloud gaming. These systems have stringent performance and scalability requirements which are often hard to meet in today's networks and using today's implementations. My research here incorporates both better understanding these applications in different settings through new measurement approaches, as well as, improving their performance and scalability through novel system designs for the underlying infrastructure driving these systems today.

Enabling Passive Measurement of Zoom Performance. To better understand and manage video-conferencing applications, we need effective ways to measure their performance in the wild to aid operators in capacity planning, troubleshooting, and setting QoS policies. Previous large-scale measurements of these applications rely on end-host cooperation, which is impractical for network operators, while in-depth analysis of captured packets requires knowledge of the packet formats. Zoom is one of the most sophisticated and popular applications, but it uses a proprietary, previously not understood network protocol. In this work [4], we were the first to demystify how Zoom works at the packet level by analyzing a previously unknown binary packet format, and propose techniques for analyzing Zoom performance solely from passively captured packets, revealing important metrics like media bit rates, frame rates, latency, and jitter. Finally, we provide unprecedented, at-scale insights into Zoom's real-world operation using packet-level data for every call on the entire Princeton campus. Our study shows a long tail of poor meeting performance across several key quality metrics, such as jitter and bit rate. All software artifacts are available for the community and have been adopted by industry and academics to study Zoom and other video-conferencing systems in the wild.

Redesigning Video-Conferencing Systems for Scale. Our measurements of Zoom and other applications have revealed that the architecture of video-conferencing systems today is not well suited for the current and expected growth of these applications, which require an unprecedented scale of their complex underlying infrastructure. This, in particular, concerns the servers that relay media among participants which must sustain high data rates and dynamically scale compute and networking resources up and down while also supporting complex logic for rapid adaptation of media streams. We found that this workload is inherently hard to scale as the amount of packets to process grows quadratically with meeting size and under-provisioning of these servers has direct impact on QoE. Our analysis of Zoom, however, also revealed that forwarding and adapting media signals in video conferencing is strikingly similar to traditional packet processing. Based on this insight and following a long line of work on software-defined networking (SDN), we introduce Scallop, a rearchitected video-conferencing system that offloads all media forwarding to an efficient *data plane*. For this, we map the complex application-layer multicast and media-adaptation logic to modern hardware-based processing engines and present a prototype providing a $7\text{-}422\times$ scaling improvement over a 32-core commodity server [5]. Our prototype also lowers media forwarding latency by $26\times$ in comparison to software which are key quality metrics for real-time applications.

Measuring and Understanding Wireless Impact on Video Conferencing and Beyond. Rapid delay variations in today's wireless networks impair the QoE of low-latency, interactive applications, such as video conferencing. Our measurements have revealed that applications essentially operate in the dark as it is hard to differentiate between short-term delay artifacts stemming from wireless scheduling and retransmissions and those caused by congestion. To tackle this problem, we propose Athena [7], a framework that correlates high-resolution measurements across Layer 1 to Layer 7 to remove the fog from the window through which today's congestion-control algorithms see the network. Athena extracts 5G control-channel telemetry of physical-layer data units, retransmissions, and scheduling decisions and precisely time-synchronizes these measurements with packets at the network layer and video frames or audio samples at the application layer. This new cross-layer view paves the way for new works studying the precise interaction between networks and real-time communication applications. The first such work atop of Athena is Domino [8], a novel root-cause analysis tool that explains precisely how 5G causes various types of QoE degradations (e.g., low resolution or video freezes) in video conferencing. Athena's cross-layer view of the network and Domino's analysis capabilities provide the networking community critical insights to revisit and re-evaluate today's scheduling, congestion-control, and RTC rate-adaptation algorithms.

Linux Networking and Acceleration of Network Functions

In other works [2, 3, 5, 6], I have accelerated different packet-processing and network-management tasks using hardware offload. Yet, we need general approaches for accelerating a wide range of network functions automatically and transparently. Linux is the most commonly used platform for software packet processing but supporting today’s required performance with its out-of-the-box networking stack is challenging.

Transparently Accelerating Linux Networking. In my work on software-based packet processing, building on years of research on high-performance software data planes, we introduce transparent acceleration into the Linux networking stack [1]. Rather than using kernel-bypass technologies, which require re-implementation of protocols and control-plane interfaces, we leverage eBPF in a novel way, as an explicit fast path for common packet-processing tasks, such as routing, or packet filtering. We integrate this fast path with the kernel’s existing control plane, unifying state between fast and slow paths. In this innovative design, standard command-line tools, control-plane software, and frameworks such as Kubernetes still work without modification and benefit from a faster network data plane, improving throughput by up to 77%.

Real-Time Network Telemetry and Control

My work on network telemetry focuses on the question of whether we can obtain and effectively leverage insights gathered from hundreds of millions of packets per second in today’s networks. This is important for security applications but also enables novel congestion-control algorithms. By taking a holistic approach that partitions measurement and analytics between hardware and software, we show that this goal is achievable.

Packet-Level Network Telemetry and Analytics with *Flow and Jetstream. My first line of work in this area explored how telemetry systems can harness the performance of programmable line-rate switches while also meeting requirements for practical deployment. We introduce *Flow [3], a highly-efficient telemetry system that supports concurrent measurement and dynamic queries through carefully partitioning between hardware and software. The pipeline runs at 3.2 Tbit/s and exports a stream of telemetry records that provides unprecedented packet-level insight into the network, e.g., for intrusion detection or network debugging. *Flow is accompanied by Jetstream, a software packet-analytics engine which processes up to 250 million packets per second on a commodity server, a 100× improvement over previous systems [6].

PowerTCP: Telemetry-Assisted Network Congestion Control. Especially in the context of data centers where congestion-control algorithms must act on sub-millisecond time scales (e.g., due to frequent micro bursts), leveraging telemetry is a promising way to provide insight into the queue state and current network congestion. In PowerTCP [2], we designed an innovative end-host congestion-control algorithm that, in contrast to the state of the art, operates on both network state (e.g., queue length and RTT) and network variation (e.g., RTT gradient) with a data-plane component that provides telemetry about bottleneck queue occupancy in real time to TCP senders. PowerTCP is able to utilize available bandwidth within one or two RTTs while being stable, maintaining low queue lengths, and resolving congestion rapidly, making it particularly suitable for reconfigurable data-center networks where the link capacity changes frequently.

Future Research Directions

I am excited to pursue more research in the future that is motivated by the challenges modern networked applications face. My measurement-driven approach has uncovered various complex and often unintended ways networks and applications interact, such as the impact of latency artifacts from access networks on video-conferencing applications [4, 7, 8]. Secondly, I have shown that cross-layer designs and optimizations are effective in mitigating many of the challenges that applications face today [2, 5].

I plan to leverage these insights to work on pressing issues such as building networks that provide the best possible service to the applications they carry. We live in an exciting time where this goal is more achievable than ever through network programmability: the network is no longer a black box but can be programmed and optimized, for example using programmable switches or programmability inside the cellular radio-access and core networks. Currently, most of my efforts focus on wireless networks; the underlying concepts, however, are more broadly applicable. Furthermore, I plan to pursue forward-looking work on future network designs that expose the right abstractions for cross-layer information sharing and optimization, with the goal of

ultimately improving applications such as real-time communication (RTC), extended reality (XR), but also, for example, distributed learning. Below, I outline three themes of my future research.

Scalable Application Performance Measurements. A key obstacle to understanding the performance and behavior of RTC applications in access networks is the complexity of measurement setups. Our previous measurement setups [7, 8] are limited in scale in that they cannot easily be deployed on Internet-wide measurement infrastructure, nor can it be run by untrained users to gauge the performance of their networks. I plan to build a measurement framework that generates traffic precisely following the patterns of RTC traffic with receivers that run the same algorithms for rate adaptation and jitter-buffer management as the real applications. Using this approach we can still give precise estimates of what quality an RTC application would offer in terms of resolution, frame rates, mouth-to-ear delay, and stall probability without running an actual video call. Systems like speedtest.net only measure raw throughput and average latency but do not provide insights into fine-grained latency, achievable media bit rates, and more broadly the QoE a user can expect. Finally, this system is lightweight enough to deploy on measurement infrastructure such as CAIDA Ark that provides probes around the globe, enabling RTC quality measurements at unprecedented scale.

Networks optimized for RTC Applications. Despite advances in 5G technology, due to rapidly fluctuating capacity and latency, the QoE of RTC applications in 5G often still remains unsatisfactory [8]. These problems are especially pronounced during congestion, when the RAN must dynamically reallocate capacity. While 5G Ultra-Reliable Low-Latency Communications (URLLC) aims to address such challenges, it is (a) far from widely deployed and (b) highly inefficient due to high spectrum consumption when applied indiscriminately. Secondly, the 5G New Radio QoS framework is not aware of application semantics such as different quality layers as they are commonly used in media streams as it cannot see beyond the IP 5-tuple. As a step to make 5G networks application-aware, we propose StreamGuide, a scheduling and prioritization system that dynamically protects those sub-flows that are most important for QoE by leveraging application semantics and continuously monitoring application QoE and network state inside the base station (gNB). Under congestion and when QoE is impaired, StreamGuide prioritizes the most relevant packets (e.g., video base layers or audio data) while preserving overall fairness in scheduling with other flows.

Enabling Cross-Layer Visibility in the Network Stack. Looking further into the future, I aim to develop more principled ways to share critical information across layers in the networking stack. The interfaces between layers today allow passing data up and down the stack but are largely not designed to carry meta data relevant for decision making at adjacent layers. Consequently, we need abstractions that enable information sharing for cross-layer optimization while leaving the layering abstraction intact. I plan to collaborate with industry partners and standardization bodies to design future-proof abstractions that are suitable for a wide range of use cases. In particular, I am working on an interface to share estimated near-future cellular capacity estimates with UEs to directly guide rate control, allowing applications to adapt rapidly without incurring queuing due to slow reaction from a traditional congestion-control loop.

References

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